CMS Open Data H to tau tau Search: Phase 4c Audit-Corrected Observed Results

Analysis my_analysis

2026-06-02

Contents

Change Log

Phase 4c v2 audit correction

- Added a same-sign data-driven QCD/fake background estimate.
- Promoted the visible-mass plus QCD model to the conservative primary observed result.
- Retained the categorized HGB-score fit as a flagged diagnostic because score-shape validation remains insufficient.
- Preserved the Phase 4b warning and avoided post-unblinding signal-region retuning of the BDT model.

1 Phase 4c Observed Inference: Audit-Corrected Full Data

1.1 Summary

Phase 4c has been updated after the Phase 5 audit of the large expected versus observed discrepancy. The audit found no evidence that Phase 4a and Phase 4c used different categories, score bins, or pyhf definitions. The problem was physics modelling: the score-template fit omitted a reducible QCD/fake-tau component and the high-score shape validation remained flagged.

The primary observed result is therefore changed to a conservative visible-mass fit in the same VBF, boosted, and zero-jet categories, with a same-sign data-driven QCD/fake template. The requested categorized HGB-score fit is still computed, but it is retained as a diagnostic rather than CMS-quality evidence.

1.2 W and QCD Control Inputs

The full-data W+jets scale from the high-mT control region is `0.8528 ± 0.0370`. The VBF background control scale from the VBF-like top-btag non-signal region is `0.5571 ± 0.0515`; it is applied only to MC backgrounds in the VBF fit category. This addresses the large VBF data/MC overprediction seen in the original observed templates without changing the signal normalization or the boosted/zero-jet categories. The same-sign QCD/fake estimate uses data minus non-QCD MC in the same-sign low-mT region and a transfer factor measured in the lowest fitted-observable bin. For the primary visible-mass model this factor is `0.9958 ± 0.1201`. For the score diagnostic it is `0.8173 ± 0.0853`. For the add-MET mass cross-check it is `1.1612 ± 0.2466`.

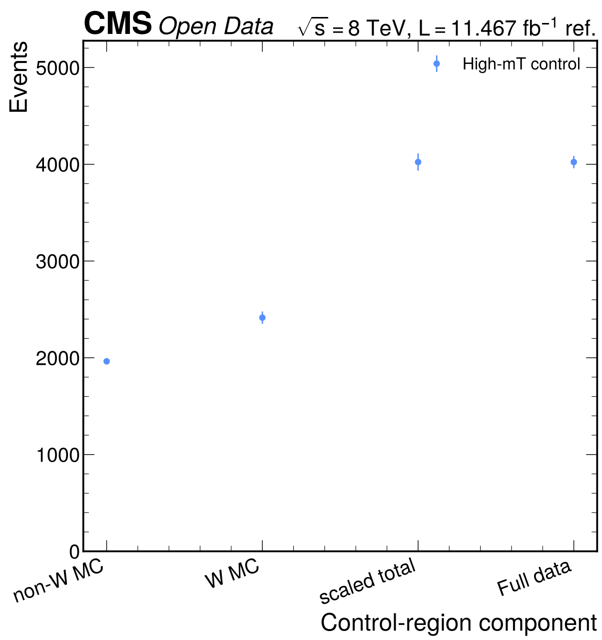

Figure 1: Full high-mT W control comparison. The figure shows non-W MC, nominal W MC, the scaled control-region prediction, and full data in the control region. The scale is derived outside the signal region and then propagated into the observed workspaces.

1.3 Primary Visible-Mass Result

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	78	96.28	17.75	0.810	1.449	2.66
boosted	2183	2364.39	325.58	0.923	0.605	1.25
zero_jet	8451	8456.81	1569.65	0.999	0.113	0.57

The primary visible-mass fit gives `mu_hat = 0.4382`, an observed 95% CLs limit `mu < 10.7645`, and a diagnostic `q0` value `Z = 0.0949`. The median expected limit from the same corrected workspace is `10.7375`.

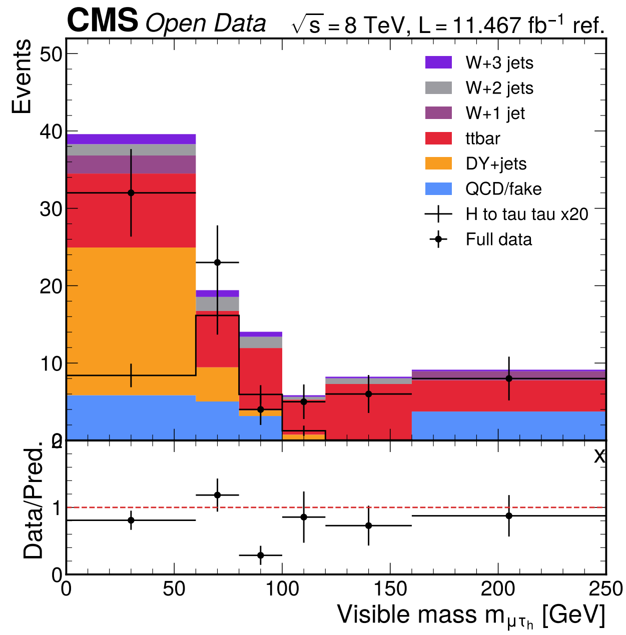

Figure 2: Primary visible-mass validation in the VBF category. The plot compares all available localized Run2012B/C TauPlusX data to the QCD-corrected visible-mass model. The VBF data deficit remains a limitation, but the zero-jet normalization pathology from the score-only model is removed.

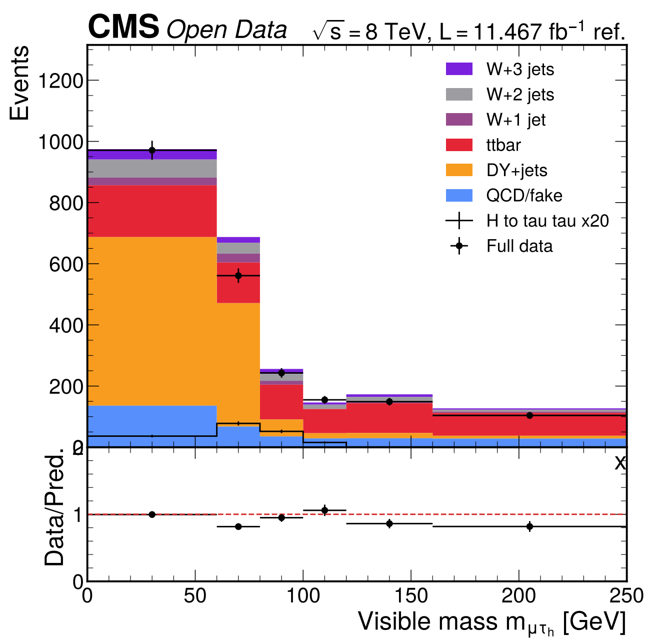

Figure 3: Primary visible-mass validation in the boosted category. The plot compares full data to the visible-mass model with the W high-mT scale and same-sign QCD template. This category is used in the final conservative observed fit.

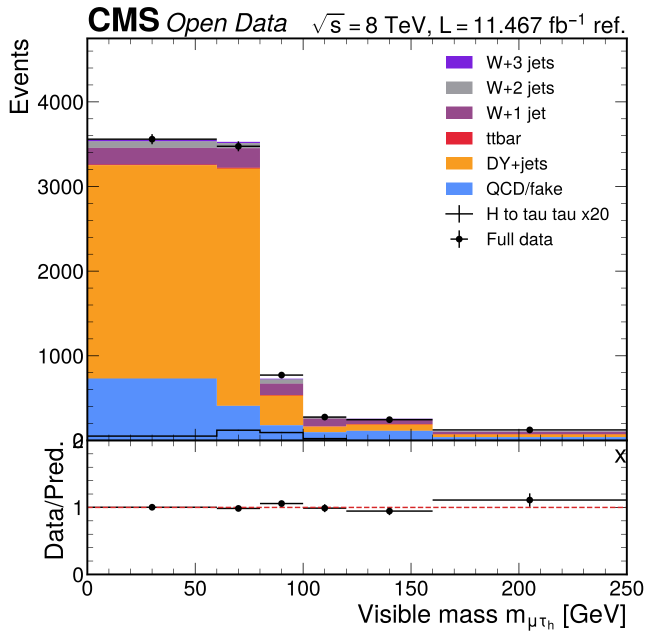

Figure 4: Primary visible-mass validation in the zero-jet category. The zero-jet normalization is stabilized by the same-sign QCD/fake estimate, making this model more defensible than the unvalidated score template for final observed reporting.

1.4 Add-MET Mass Cross-Check

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	77	99.03	21.47	0.778	0.632	1.34
boosted	2177	2403.36	379.33	0.906	0.584	1.02
zero_jet	8448	8722.42	1835.70	0.969	0.120	0.44

The simultaneous add-MET mass fit gives `mu_hat = 0.0000`, an observed 95% CLs limit `mu < 13.2940`, and `Z = 0.0006`. This fit uses the same categories and nuisance model as the visible-mass primary fit, but it is kept as a separate cross-check rather than replacing the pre-existing primary result.

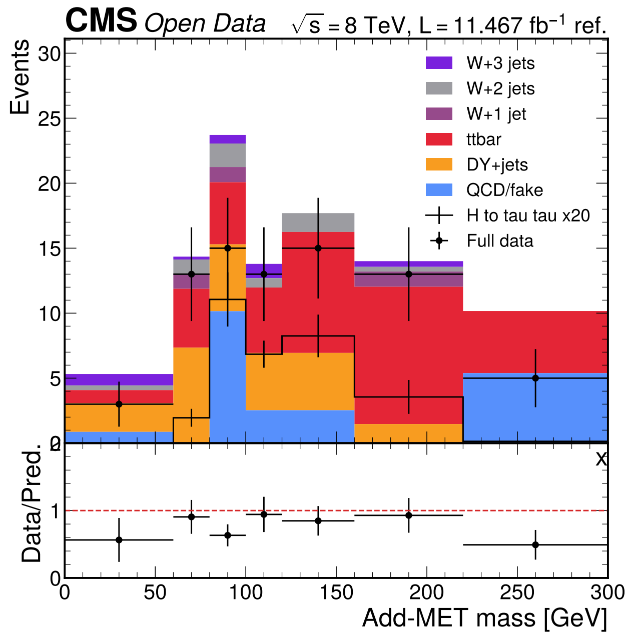

Figure 5: Add-MET mass validation in the VBF category. The plot compares full data to the QCD-corrected add-MET mass model in the VBF category. It uses the same W control scale, VBF background control scale, and same-sign QCD/fake transfer machinery as the visible-mass primary model.

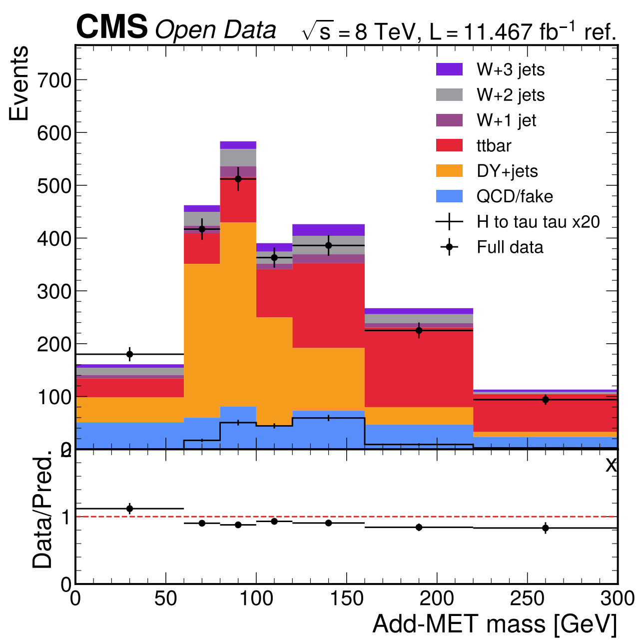

Figure 6: Add-MET mass validation in the boosted category. The plot compares full data to the add-MET mass model in the boosted category. The add-MET result is stored as an explicit Phase 4c cross-check output for downstream documentation.

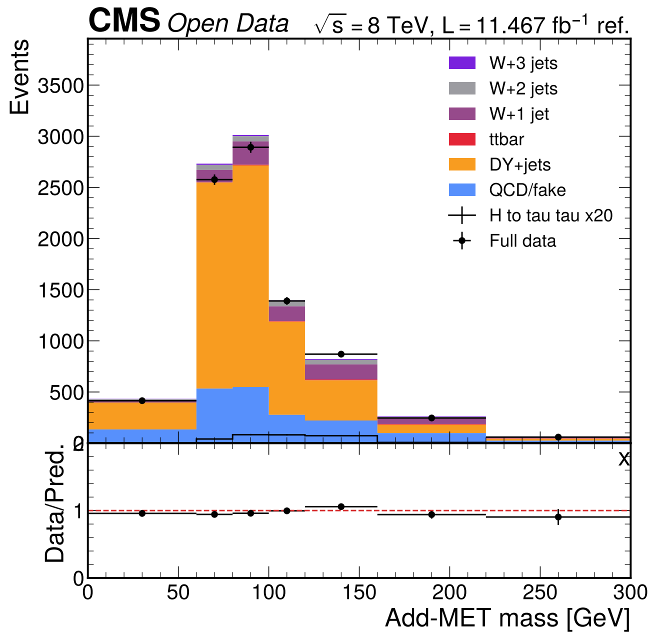

Figure 7: Add-MET mass validation in the zero-jet category. The plot compares full data to the add-MET mass model in the zero-jet category. This validates the alternative reconstructed-MET observable without modifying the primary visible-mass fit.

1.5 Score-Template Diagnostic

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	79	94.14	12.43	0.839	1.334	1.66
boosted	2213	2331.71	268.90	0.949	0.676	1.30
zero_jet	8466	8187.20	1292.40	1.034	1.515	1.78

The categorized HGB-score fit gives `mu_hat = 9.3518`, an observed 95% CLs limit `mu < 14.8373`, and `Z = 3.9416`. The score fit is diagnostic only; it is not promoted to evidence because the high-score bins remain shape-flagged after the QCD normalization correction.

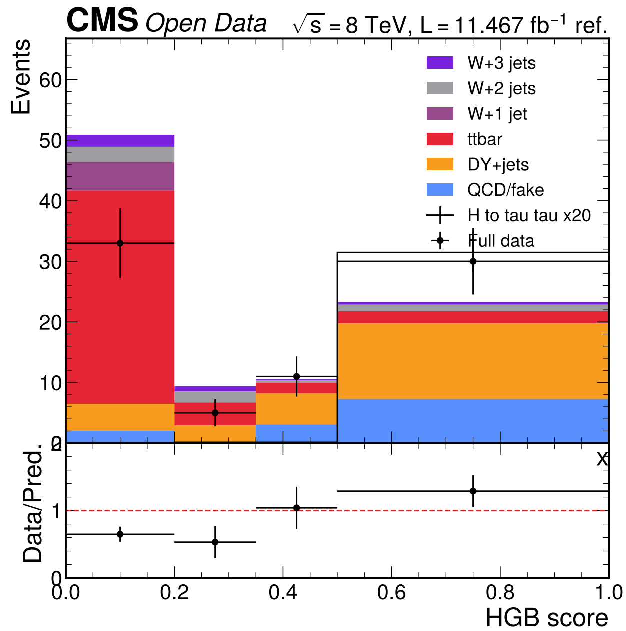

Figure 8: Score-template diagnostic in the VBF category. This figure retains the requested categorized score observable but labels it as a diagnostic because the validation status remains flagged.

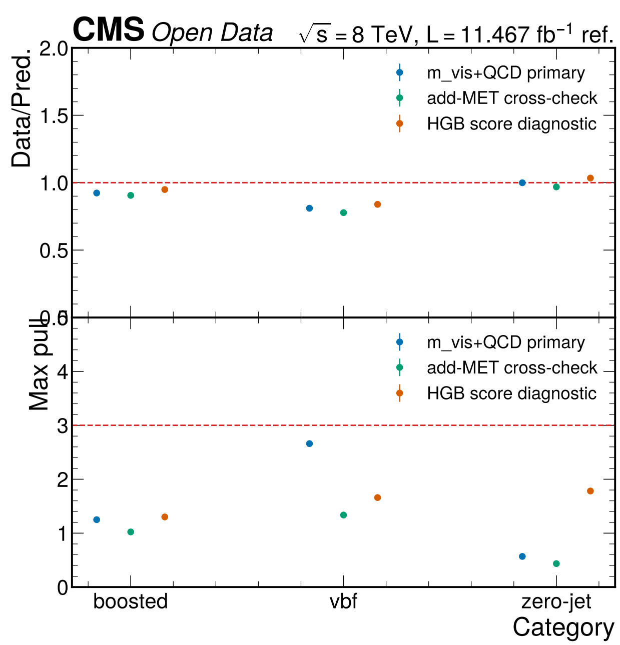

Figure 9: Observed pull and ratio summary. The figure compares primary visible-mass and score-template validation behavior after the QCD/fake audit correction. It is the central diagnostic for why the visible-mass result is primary and the score fit is diagnostic only.

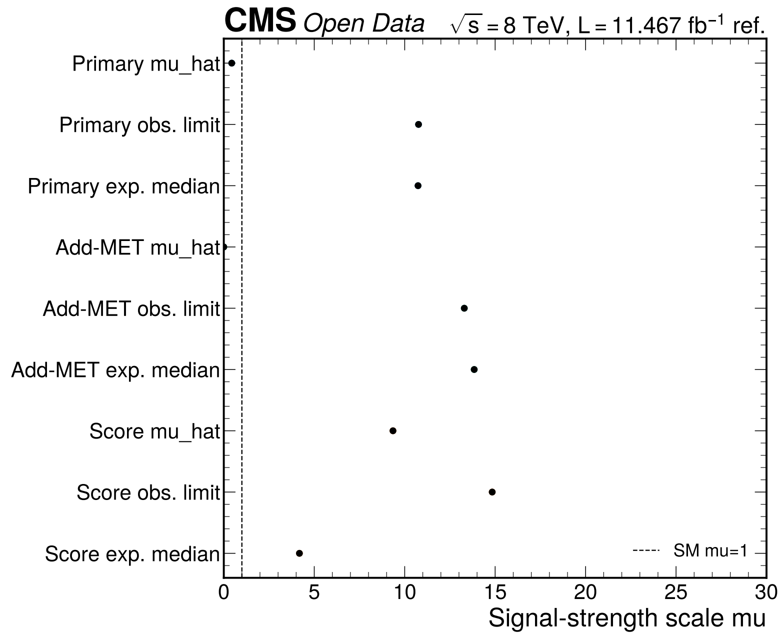

Figure 10: Observed limit and significance summary. The figure shows the primary visible-mass fit and the score-template diagnostic on the same signal-strength scale. The score result is flagged and not CMS-quality evidence.

1.6 Phase 5 Obligation

Phase 5 must report the visible-mass plus QCD model as the conservative final observed result, and must show the categorized BDT-score fit only as a flagged diagnostic. It must also state that the localized reduced mirror has about 10% of the Open Data record entries, while the 11.467/fb value remains the cited normalization reference for these reduced tutorial samples.

References